



Figure 3: Edit success for forward and reverse multiple-choice test queries by editor type. Dashed line indicates chance performance for forward and reverse edits by color.

Imagine that a Martin D-28 is a kind of trumpet.

Question: a Martin D-28 has _____

A: three valves
 B: 17 pads
 C: six strings
 D: 88 keys
 E:

Please choose the best answer based on the instructions to imagine.

A

B

C

D

E

Figure 4: Screenshot illustrating a trial of the of human annotation study.

tator agreed with the "correct" answers on 95/100 ratings. Overall, human behavioral data indicate that the task prescribed by TAXI is human solvable and set a benchmark that far exceeds any existing editors' performance.

	Subjects			Properties
Dog	Labrador,	Chihuahua,	Pekingese, Bichon Frise	Born in a litter, has fur, is domesticated, has four legs, kept as pets, likes to fetch, barks, walks, baby is a puppy, is a mammal
Cat	Siamese,	Persian,	Abyssinian, Chartreux	Born in a litter, has claws, has fur, is domesticated, has four legs, kept as pets, likes to chase, meows, walks, baby is a kitten, eats meat
Cow	Holstein,	Jersey,	Gal-loway, Hereford	Is a mammal, born alone, has hooves, has fur, is domesticated, has four legs, kept for their milk, likes to graze, moos, walks, baby is a calf, eats grain, makes milk
Bird	sparrow,	canary,	wood-pecker, Partridge	Is an aves, hatched from egg, has wings, has feathers, chirps, flies, baby is a chick, can fly, is wild
Bee	Bumblebee,	Honeybee,	Megachile, Apis Mellifera	Is an insect, has wings, has six legs, buzzes, flies, makes honey, can fly
Fish	Trout,	salmon,	Flounder, tilapia	Hatched from an egg, has scales, has fins, has no legs, caught and eaten, swims, lives in water, can swim, is wild
Snake	Cobra,	python,	copper-head, Gaboon viper	Hatched from an egg, has scales, no legs, people avoid, hisses, slithers, is wild

Table 5: Taxonomic details for superordinate category "animals" from the TAXI benchmark dataset. The first two listed exemplars are typical exemplars.